Final Results

PDF edition of preserved repository document | docs/final_results.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Final results

Final main study:90 adapted heads (84 primary plus 6 retrained ablations),210 shared-runner evaluations, three seeds 0/1/2,200 IID and300 OOD examples per run. An additional12 historical MLP models/24 evaluations and6 non-neural reference evaluations use the same fresh final dataset. Total final research evaluations240;102 trained research models. Diagnostics are separate. Raw results remain in results/final/; the committed summary snapshot is experiments/final_smollm2_v1/.

Primary results at 1000 training problems

Condition	IID exact accuracy	Structural OOD exact accuracy	IID groundedness	IID goal success	IID trajectory validity
Answer-only A	5.000% ± 0.866	0.000% ± 0.000	0.000% ± 0.000	0.000% ± 0.000	N/A
Structured baseline	5.333% ± 1.041	0.000% ± 0.000	0.000% ± 0.000	0.000% ± 0.000	N/A
Mindscape B	41.500% ± 2.179	6.111% ± 1.347	41.500% ± 2.179	41.500% ± 2.179	41.500% ± 2.179
Mindscape C	19.167% ± 3.014	0.000% ± 0.000	19.167% ± 3.014	19.167% ± 3.014	19.167% ± 3.014

All rates are mean±sample SD across three training seeds. A/structured do not emit trajectories; validity isN/A and their evidence-required grounding/goal scores are0 by protocol. B OOD groundedness/goal/validity are6.111%±1.347; C OOD rates are0. No final success target was met.

Learning curves

IID

Training problems	Answer-only A	Structured	B	C
10	0.000% ± 0.000	0.000% ± 0.000	0.000% ± 0.000	0.000% ± 0.000
25	0.167% ± 0.289	0.000% ± 0.000	2.000% ± 0.000	0.500% ± 0.866
50	0.333% ± 0.289	0.167% ± 0.289	3.000% ± 1.323	0.500% ± 0.500
100	0.667% ± 0.289	0.667% ± 0.289	13.333% ± 1.041	2.667% ± 2.517
250	2.333% ± 0.577	0.833% ± 0.577	30.333% ± 1.041	9.667% ± 3.215

Training problems	Answer-only A	Structured	B	C
500	1.333% ± 0.764	2.333% ± 0.764	38.000% ± 0.866	15.000% ± 3.606
1000	5.000% ± 0.866	5.333% ± 1.041	41.500% ± 2.179	19.167% ± 3.014

Structural OOD

Training problems	Answer-only A	Structured	B	C
10	0.000% ± 0.000	0.000% ± 0.000	0.000% ± 0.000	0.000% ± 0.000
25	0.000% ± 0.000	0.000% ± 0.000	0.111% ± 0.192	0.000% ± 0.000
50	0.000% ± 0.000	0.000% ± 0.000	0.000% ± 0.000	0.000% ± 0.000
100	0.000% ± 0.000	0.000% ± 0.000	0.889% ± 0.192	0.000% ± 0.000
250	0.000% ± 0.000	0.000% ± 0.000	1.556% ± 0.694	0.111% ± 0.192
500	0.000% ± 0.000	0.000% ± 0.000	4.667% ± 0.333	0.111% ± 0.192
1000	0.000% ± 0.000	0.000% ± 0.000	6.111% ± 1.347	0.000% ± 0.000

Data efficiency and costs

All observed N*(0.8/0.9/0.95) thresholds were not reached on either split for every condition; all DERs are censored/not reached. There is no measured2× or5× threshold-crossing advantage. Problem budget is distinct from label, feedback-query and compute budget.

Main study measured training time 171.947s and evaluation time 886.511s. It performed108,000 optimizer updates,6,912,000 sampled supervision rows,379,153 scalar feedback queries and303,924 prediction-head calls. The shared frozen transformer processed1,961,882 tokens in23,463 batches; encoder feature requests726,352. These are cached CPU measurements, exclude dataset/artifact bookkeeping outside timed sections, and do not establish cold-deployment compute efficiency. Full per-model costs are in costs.json.

Ablations at 50 problems

Intervention	IID Δaccuracy (pp)	OOD Δaccuracy (pp)	IID Δgroundedness (pp)	IID Δgoal (pp)
dream	0.000	0.000	0.000	0.000
matched_information_control	0.000	0.000	0.000	0.000
no_dream	0.000	0.000	0.000	0.000

Intervention	IID Δaccuracy (pp)	OOD Δaccuracy (pp)	IID Δgroundedness (pp)	IID Δgoal (pp)
no_environment_interaction	0.000	0.000	-3.000	-3.000
no_episodic_memory	0.000	0.000	0.000	0.000
no_explicit_state	-3.000	0.000	-3.000	-3.000
no_goal_evaluator	0.000	0.000	0.000	0.000
no_trajectory_supervision_regime_C	-2.500	0.000	-2.500	-2.500

Every intervention has three seeds. Δdata efficiency is not estimable from one budget. No goal evaluator means removal of a constant goal input flag while retaining common external scoring. C versus B is a supervision/feedback contrast, not an isolated removal. No real interaction preserves numerical predictions but loses real evidence by definition. The matched local supervised policy and matched-order/update no-SQLite-retrieval control have identical predictions to their references. Dream changed no measured accuracy/groundedness/goal; its extra calls do not support usefulness. The no-state mean reduction is not significant in every seed after the component-test Holm family, so state-contribution claims remain inconclusive at this budget.

Statistics and errors

Per-run Wilson95 intervals report conditional example uncertainty; seed SD reports training variation. Exploratory paired tests use seed 0 at 1000, six Holm comparisons; component tests are a separate 18-comparison family across seeds/splits. No seed pooling. B versus A IID Holm p=2.2354e-17 and OOD p=1.90735e-6; C versus A IID p=1.80444e-7. These differences are observed regime associations, not controls for the unequal information/scaffold. Practical OOD performance remains6.11%, far below90%.

Failure/outcome category	Count across210 repeated-set evaluations
arithmetic_error	22367
wrong_action	28762
invalid_transition	0
malformed_state	0
trajectory_failure	0
goal_failure	0
unsupported_answer	133
timeout	0
other	0
correct	1238

These counts describe repeated benchmark runs, not unique independent errors. difficulty_failures.json groups condition/budget/seed/IID-OOD/digit structure/carry count, and per-run failures.json retains example

identities and original taxonomy. Main failures are wrong local claims and answer arithmetic errors; syntax/timeouts were not the limiting issue. At1000,108/200 IID examples and280/300 OOD examples contain reference local contexts absent from the training subset. Carry signatures are unseen for59/200 IID and300/300 OOD. The latter includes the structural length change; these are posthoc descriptive strata, not independently pre-reserved carry-only holdouts.

Sanity and interpretive references

The final supervised representation memorized n=10, all 45 canonical1x1 problems and a mixed100 training set at 100% in diagnostics. Mixed100 validation reached3% answer-only and18% B; local claim accuracy62.47%, carry label accuracy82.60%. C n=10 same-set accuracy was0% with incomplete accepted experience; it did not receive complete trajectories. This separates fitting ability from generalization and exploration coverage.

Historical MLP on fresh final dataset at 1000	IID mean±SD	OOD mean±SD
Answer-only A	0.333% ± 0.289	0.000% ± 0.000
Structured baseline	0.333% ± 0.289	0.000% ± 0.000
Mindscape B	8.000% ± 0.866	0.667% ± 0.000
Mindscape C	8.333% ± 2.021	0.333% ± 0.333

The original phase-4 dataset remains separate: its OOD accuracy was0% across all primary runs. Different sampled datasets can change rare correct counts. Larger final heads/categorical features/active losses/update budgets/pretrained features changed together, so gains over the MLP do not isolate pretraining. Deterministic execution scored 100% IID/OOD; operand memorization and the seeded uniform-integer random reference scored 0%. The deterministic ceiling is an algorithm, not a learning/data-efficiency baseline.

Claims

Claim	Status	Evidence	Limits
Mindscape improves accuracy	INCONCLUSIVE	See primary curves and matched-information control.	State/scaffold and supervision differ from answer baselines; wrapper causal benefit must exceed matched local policy.
Mindscape improves data efficiency	NOT SUPPORTED	Observed N* and DER only.	Unequal label/query counts; compute reported separately.
Mindscape improves structural OOD generalization	INCONCLUSIVE	Unseen structures evaluated separately.	One domain and hand-defined cursor.
Mindscape improves groundedness	INCONCLUSIVE	Verified executed evidence measured.	Answer-only protocol emits no trajectory; metric favors evidence emitters by definition.

Claim	Status	Evidence	Limits
Mindscape improves trajectory validity	INCONCLUSIVE	Complete verifier checks.	Baselines without trajectories have null validity; matched policy is required.
Explicit state contributes	INCONCLUSIVE	Retrained no-state intervention at n=50.	One budget and changes representation. Support, if present, is limited to IID at n=50; OOD is separately reported.
Episodic memory contributes	NOT SUPPORTED	Matched accepted rows/order/updates no-storage control.	Tests SQLite retrieval versus in-memory replay, not all reuse.
Dream simulation contributes	NOT SUPPORTED	Depth2/branch3 intervention n=50.	Confidence scoring, not learned world model; extra inference calls.
Architecture transfers to another environment	NOT SUPPORTED	Goal-directed generic integer dream demo retained.	No second learned benchmark; extension deferred.

Supported descriptive findings: under these tested unequal-supervision protocols, B/C have higher IID accuracy than answer-only, and B executes some verified OOD trajectories. These observations do not establish an independent Mindscape wrapper advantage; matched-information policy predictions are identical. Data-efficiency, replay-storage, dream usefulness and second-domain-transfer claims are not supported. Broader causal claims remain inconclusive.

Artifact audit:438 actual checks passed, including nested training IDs, shared parameter count, no singleton forcing, proposal/result separation, posthoc judgments, hypothetical tags and matched-control prediction equality.

Figures

- accuracy (../experiments/final_smollm2_v1/plots/accuracy.svg)

- ood_accuracy (../experiments/final_smollm2_v1/plots/ood_accuracy.svg)

- groundedness (../experiments/final_smollm2_v1/plots/groundedness.svg)

- goal_success (../experiments/final_smollm2_v1/plots/goal_success.svg)

- trajectory_validity (../experiments/final_smollm2_v1/plots/trajectory_validity.svg)

- data_efficiency (../experiments/final_smollm2_v1/plots/data_efficiency.svg)

- ablation_effects (../experiments/final_smollm2_v1/plots/ablation_effects.svg)

- errors (../experiments/final_smollm2_v1/plots/errors.svg)

- training_compute (../experiments/final_smollm2_v1/plots/training_compute.svg)

- inference_cost (../experiments/final_smollm2_v1/plots/inference_cost.svg)

<!-- emergency-mps-results -->

Mindscape bounded MPS completion evidence

Learning curve: 1920/1,920 episodes, 0 not run. Lockbox: 400/400 condition-task episodes (100 independent tasks). Preserved CPU work remains intact.

All metrics below derive from saved episodes. Missing cells are not zero scores. CPU float32 and MPS reduced-precision strata are kept separate. Supervision and interaction budgets differ across conditions; architecture-only causality is not established.

Condition	Split	Completed/planned	Success	95% Wilson interval
model_only	test	50/50	16.0%	[0.0833742067803340 4, 0.285142160630349 9]
model_only	ood_test	50/50	24.0%	[0.1429739139699173, 0.3741268375794292]
structured	test	50/50	100.0%	[0.9286524008666414, 1.0]
structured	ood_test	50/50	44.0%	[0.3116219921125717, 0.5769397197834315]
mindscape_b	test	50/50	90.0%	[0.7863976856252035, 0.9565242350681096]
mindscape_b	ood_test	50/50	48.0%	[0.3479713528657804 6, 0.614882551099553 9]
mindscape_c	test	50/50	98.0%	[0.8950455641036219, 0.9964607407283539]
mindscape_c	ood_test	50/50	36.0%	[0.2413874965184674 1, 0.498589831238873 07]

Observed complete-cell sample thresholds and all partial cells are in results/coding/emergency_analysis_v1/summary.json. DER is undefined for a causal architecture comparison. No zero-budget control was run.

Task-paired confidence intervals are descriptive and unadjusted for multiple comparisons. Transition validity is conditional on recorded actions; malformed model proposals are reported separately.

Observed self-correction from saved offline first-edit assessments: [{"condition": "mindscape_b", "split": "test", "completed_episodes": 50, " first_edit_failures": 0, "later_terminal_successes": 0, "conditional_self_correction_rate": null, "wilson_ci95": null, "definition": "Terminal success after an incorrect first accepted edit; first-edit assessment occurs offline after policy termination, never as hidden model feedback", "caution": "Descriptive within-episode self-correction; no injected-fault recovery intervention or causal memory/planner attribution"}, {"condition": "mindscape_b", "split": "ood_test", "completed_episodes": 50, "first_edit_failures": 38, " later_terminal_successes": 12, "conditional_self_correction_rate": 0.3157894736842105, "wilson_ci95": [0.1908460367559618,

0.4745575989060954], "definition": "Terminal success after an incorrect first accepted edit; first-edit assessment occurs offline after policy termination, never as hidden model feedback", "caution": "Descriptive within-episode self-correction; no injected-fault recovery intervention or causal memory/planner attribution"}, {"condition": "mindscape_c", "split": "test", "completed_episodes": 50, " first_edit_failures": 1, "later_terminal_successes": 0, "conditional_self_correction_rate": 0.0, "wilson_ci95": [0.0, 0.7934506856227626], " definition": "Terminal success after an incorrect first accepted edit; first-edit assessment occurs offline after policy termination, never as hidden model feedback", "caution": "Descriptive within-episode self-correction; no injected-fault recovery intervention or causal memory/planner attribution"}, {"condition": "mindscape_c", "split": "ood_test", "completed_episodes": 50, "first_edit_failures": 32, " later_terminal_successes": 0, "conditional_self_correction_rate": 0.0, "wilson_ci95": [0.0, 0.1071791982550706], "definition": "Terminal success after an incorrect first accepted edit; first-edit assessment occurs offline after policy termination, never as hidden model feedback", "caution": "Descriptive within-episode self-correction; no injected-fault recovery intervention or causal memory/planner attribution"}]. This is separate from unrun injected-fault recovery tests.

Operational answer metrics: {"definition": "Successful terminal program AND independent full-trajectory replay agreement; no semantic hallucination detector", "denominator": 400, "grounded_successful_answers": 228, "terminal_incorrect_answers": 172, "semantic_hallucination_rate": null, "semantic_hallucination_status": "Not measured by this program-repair protocol", "unverified_episodes": 0, "grounded_successful_answer_rate": 0.57, "unsupported_by_terminal_tests_rate": 0.43}. Failed terminal tests are unsupported program answers under this oracle; they are not a measured semantic hallucination rate.

Original frozen configuration SHA-256: 11d2a6b5bef6517a5cae10dca46f1e94bfe2db3504afd3942434fcb78f496bb0. Seeds: 11/23/37 for learning; seed 11 and budget 100 for lockbox. Exact checkpoint/source hashes are in the frozen protocol; host/runtime and training compute are in the machine-readable summary.

Actual acceleration measurements


```
{
  "cpu_seconds_per_episode": 111.3537078230629,
  "cpu_reference_scope": "Preserved fresh CPU float32 episodes; not rerun",
  "mps_throughput_seconds_per_episode": 27.54043339325017,
  "mps_mean_episode_wall_seconds": 372.8890333698155,
  "throughput_speedup": 4.043280882077708,
  "mps_episodes_per_hour": 130.7168971742586,
  "selected_batch_size": 16,
  "precision": "float16",
  "load_seconds": 22.024639666007715,
  "peak_process_rss_bytes": 3421274112,
  "mps_allocated_bytes": 3333364480,
  "mps_driver_bytes": 12526354432,
  "profiling_seconds": 600.09845495224,
  "interpretation": "Historical CPU vs fresh batched MPS throughput; per-case latency
reported separately. Arithmetic precision differs; learning-curve hardware strata remain
disclosed."
}
```


One phase-boundary memory release: {"before_driver_bytes": 10687807488, "after_driver_bytes": 4346019840, "live_allocated_bytes": 3112987904, "reason": "Single profiling/evaluation boundary; no per-operation cache clearing", "configuration_reused": true, "time_unix": 1791468859.762655}.

CPU reference episodes were previously measured with uncached inference; they were not rerun. MPS sec/episode is amortized throughput, distinct from per-episode latency while waiting for batches. Device tensors and model placement are checked; unsupported MPS operations fail with fallback disabled. Native allocation reported by the user is verified, but inference use is only established by native run outputs.

Replay and latency

```
{
  "replay": {
    "episodes": 400,
    "planned": 400,
    "final_outcome_agreement": 1.0,
    "structured_evidence_agreement": 1.0,
    "state_agreement": 1.0,
    "state_comparisons": 2584,
    "tool_result_agreement": 1.0,
    "tool_result_comparisons": 1092
  },
  "latency": [
    {
      "condition": "model_only",
      "episodes": 4,
      "planned": 4,
      "measurement_source": "Dedicated cache-disabled stage benchmark",
      "device_precision_counts": {
        "mps/float16": 4
      },
      "actual_batch_sizes": [
        1
      ],
      "peak_process_rss_bytes": 3802660864,
      "peak_mps_live_bytes": 3132599552,
      "peak_mps_driver_bytes": 4389109760,
      "peak_cuda_allocated_bytes": null,
      "peak_cuda_reserved_bytes": null,
      "sequential_episodes_per_hour": 1475.2094699714482,
      "percentiles": {
        "total": {
          "p50": 2.5287001245001193,
          "p95": 2.9951699749499765,
          "n": 4
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        "ttft": {
          "p50": 0.10764597949992094,
          "p95": 0.8070860983006238,
          "n": 4
        },
        "response": {
          "p50": 2.2023833544999434,
          "p95": 2.6520218813000156,
          "n": 4
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          "p50": 2.2023833544999434,
          "p95": 2.6520218813000156,
          "n": 4
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        "environment_execution": {
          "p50": 0.00013612449993161135,
          "p95": 0.0004516401001183111,
          "n": 4
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        "verification": {
          "p50": 0.3319560420000016,
          "p95": 0.3436878607001745,
          "n": 4
        },
        "test_execution": {
          "p50": 0.3318179789998794,
          "p95": 0.34356011874992876,
          "n": 4
        }
      },
      "caution": "Descriptive sample; stage timers overlap. TTFT is model first-token time
 after CPU input encoding. Response is per-call wall time including encoding/decoding in
dedicated measurements; total is multi-step episode wall time. Batched smoke model
durations are shared across requests. Unmeasured timers are not inferred."
    },
    {
```

```
      "condition": "structured",
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      "planned": 4,
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      "device_precision_counts": {
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      "peak_process_rss_bytes": 3802660864,
      "peak_mps_live_bytes": 3134778624,
      "peak_mps_driver_bytes": 4431052800,
      "peak_cuda_allocated_bytes": null,
      "peak_cuda_reserved_bytes": null,
      "sequential_episodes_per_hour": 1252.859663165493,
      "percentiles": {
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          "p50": 2.891284770499624,
          "p95": 3.226074864299926,
          "n": 4
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        "ttft": {
          "p50": 0.3468195834998369,
          "p95": 0.3796293562498249,
          "n": 4
        },
        "response": {
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          "p95": 2.895561100049872,
          "n": 4
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          "p95": 2.895561100049872,
          "n": 4
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        "environment_execution": {
          "p50": 0.0006867300003250421,
          "p95": 0.0008416026992108527,
          "n": 4
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        "verification": {
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          "p95": 0.33450531509984105,
          "n": 4
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        "test_execution": {
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          "p95": 0.3343919018002907,
          "n": 4
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      "caution": "Descriptive sample; stage timers overlap. TTFT is model first-token time
 after CPU input encoding. Response is per-call wall time including encoding/decoding in
dedicated measurements; total is multi-step episode wall time. Batched smoke model
durations are shared across requests. Unmeasured timers are not inferred."
    },
    {
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```
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          "p95": 44.30880428300006,
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          "n": 32
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        "response": {
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          "n": 32
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          "n": 4
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 after CPU input encoding. Response is per-call wall time including encoding/decoding in
dedicated measurements; total is multi-step episode wall time. Batched smoke model
durations are shared across requests. Unmeasured timers are not inferred."
    },
    {
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      "episodes": 4,
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      "peak_cuda_reserved_bytes": null,
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          "n": 4
        },
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        "verification": {
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          "p95": 0.38461931249967163,
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        "test_execution": {
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          "p95": 0.7454004454003552,
          "n": 4
        }
      },
      "caution": "Descriptive sample; stage timers overlap. TTFT is model first-token time
 after CPU input encoding. Response is per-call wall time including encoding/decoding in
dedicated measurements; total is multi-step episode wall time. Batched smoke model
durations are shared across requests. Unmeasured timers are not inferred."
    }
  ]
}
```


Preserved public benchmark

0.5B: 94/164 (57.32%); 1.5B: 98/164 (59.76%). Greedy one-sample full-module generation, 512-token cap, CPython 3.14.7 WASI. Foundation pretraining exposure is unknown. These local results do not exceed historical GPT-4 67.0% or Gemini Ultra 74.4%, and protocols differ. See docs/historical_references.md for pinned primary sources.

Dedicated latency: 16/16 fresh sequential cases; actual batch sizes and memory are reported above. Sequential response latency is distinct from frozen batch-16 evaluation throughput.

Complete suite: {'tests': 136, 'failures': 0, 'errors': 0, 'skipped': 0, 'pytest_passed_tests': 130, 'subtests_passed': 6, 'seconds': 22.683}. Local fairness/leakage assertions: 10158, PASS. Contamination-free status remains unknown. All 56 retained adapters validated. SQL environment implemented; SQL model study deferred. Recovery and memory/dream ablations not run; their claims remain inconclusive.